Lecture 6.4: Experiments, Qualitative Evidence, and Inference

AREC 570 / ECON 530 · Unit 6

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Before class

Read

- Gibson, “Are You Estimating the Right Thing? An Editor Reflects” (Gibson 2019)
- Kim, “Tackling False Positives in Business Research” (Kim 2019)

Additional readings on experiments and qualitative research

- Falk and Heckman on laboratory experiments (Falk and Heckman 2009)
- Starr on qualitative and mixed methods (Starr 2014)

Readings and course handouts are available on Canvas

Methods covered in Unit 6

- **6.1** Used reduced-form comparisons and developed an estimating equation and identifying assumption from a research question
- **6.2** Estimated structural parameters from a model of behavior and used the model to examine counterfactuals
- **6.3** Used machine learning for prediction and within other empirical analyses

6.4 covers experiments, qualitative and mixed methods, and the measurement and statistical decisions that support a conclusion

Our hypothetical example

Setting: Irrigation districts deliver surface water to member farms. In a drought year, the districts cut allocations and want farms to use less water without building new infrastructure

Problem: Farms see their water use on a seasonal bill, after most irrigation decisions are made

Proposed program: Mail each farm a monthly report of its water use, compared with similar farms nearby. Producing and mailing the reports costs the districts money

Policy question: Does offering farms a monthly water-use report reduce irrigation water use enough to justify its cost?

Related evidence on feedback reports: Allcott (2011) on household electricity use; Ferraro and Price (2013) on residential water use

Following the example through the lecture

1. **Choose a design:** Which method fits the districts' question? Field and lab experiments, interviews, and mixed methods
2. **Are we estimating the right thing?** Which farms and which measure of water use? (Gibson)
3. **Plan the sample:** How many completed surveys does the study need? (Kim)
4. **Interpret the results:** What do the data support once they arrive? (Kim)

All numbers are hypothetical

Choose a design

Which method fits the districts’ question?

Quantitative and qualitative evidence

Approach	Evidence	Research use
Quantitative	Numerical measures from surveys, records, or experiments	Estimate patterns, effects, parameters, or predictions
Qualitative	Interviews, field observations, or documents	Document decisions and the processes that produce them
Mixed methods	Numerical and qualitative evidence	Examine a question using measured outcomes and accounts of decisions

Select a method to match the research task

Research task	Evidence or method
Estimate a policy effect	Random assignment or an observational comparison with a stated identifying assumption
Estimate preferences	Choice data and a model of consumer or producer behavior
Predict a future outcome	A prediction rule evaluated with held-out observations
Document a decision process	Interviews, fieldwork, or case studies
Test a response to an incentive	A field or lab experiment

Field experiment (1/3)

The districts control who receives the report, so they can randomly assign the offer

Design element	Hypothetical study
Eligible farms	Farms on the districts' customer lists
Assignment	Random offer of the report at the farm level
Outcome	Total monthly water use in acre-feet per farm-month, collected in a follow-up survey
Effect of interest	A reduction of **2 acre-feet per farm-month** would justify the cost of the program

Field experiment (2/3)

One study with two parts: the **random offer** is the experiment, and the **follow-up survey** measures water use in both groups

1. Assign eligible farms to the offer group (treatment) or the comparison group (control)
2. Survey both groups about subsequent monthly water use
3. Compare group means to estimate the effect of **being offered** the report
4. Document report delivery, survey response, and departures from assignment

The estimate applies to the offer among farms included in the experiment. Extrapolation to other districts or a mandatory reporting program requires evidence about those settings (i.e., generalizability)

Field experiment (3/3)

The analysis subtracts mean water use in the comparison group from mean water use in the group offered the report

$$\hat{\tau} = \bar{Y}_{\text{offer}} - \bar{Y}_{\text{comparison}}$$

A negative $\hat{\tau}$ indicates lower reported use among farms offered the report. With comparable follow-up in both assigned groups, the difference estimates the effect of the offer

Before computing $\hat{\tau}$, the team must decide what it measures, for which farms, and how many completed surveys it requires

Lab experiments

Falk and Heckman's gift-exchange experiment examines whether workers respond to higher wages with higher effort

1. Participants act as firms or workers
2. A firm offers a wage; a worker who accepts chooses an effort level that is costly to provide
3. The experiment specifies the available choices, minimum effort, and payoff rules

What the lab adds: Control over choices, information, and payoffs, so researchers can vary one feature at a time and replicate the design

What it leaves out: The field setting and the population named in the policy question. Falk and Heckman argue that generalizability is a question for any single study, lab or field

Source: Falk and Heckman (2009), discussion of the gift-exchange experiment

Interviews

Research use: Interviews can document whether farms received and understood the report, how they used it, and which irrigation changes were feasible

Study procedures

- Recruit across farm size, crop, water rights, and location
- Document the interview guide and follow-up questions
- Describe recording, transcription, and coding
- Examine accounts that challenge the proposed explanation

Reporting: State the recruitment criteria and evidence for each interpretation

Source: Starr (2014), Sections 2 and 3

Mixed methods designs

Interviews (qualitative) combined with the follow-up survey (quantitative) in different sequences

Sequence in the water-use study	Contribution
Interviews before the survey	Identify how farms track water use and refine the survey measure
Interviews after the survey	Investigate why some farms used the report and others did not
Concurrent interviews	Compare accounts of irrigation changes with reported water use

The interviews examine how farms responded to the report. The survey comparison estimates the effect of offering it

Source: Starr (2014), Section 2

Are we estimating the right thing?

Which farms do the data cover, and which measure of water use does the survey record?

Four questions to ask of any estimate

Who is covered?

- Population: Units of interest
- Example: Farms on district customer lists

What was recorded?

- Measure: Recorded indicator of a concept
- Example: Survey-reported water use

Compared with whom?

- Comparison: Groups whose outcomes are compared
- Example: Farms offered a report and farms not offered one

What narrows the claim?

- Limits: Conditions that restrict the conclusion
- Example: Missing survey responses or unlisted farms

Source: Gibson (2019)

Population represented in the data

- **Included:** Farms served by participating districts appear on the customer lists
- **Possibly excluded:** Farms relying entirely on private wells or other suppliers may be absent
- **Population claim:** An estimate for farms on the lists requires additional evidence before applying it to all farms in the region

Gibson's distinction between **registered** and **resident** households illustrates how a data source can omit members of the population named in the research question

Source: Gibson (2019), pages 331 to 332

Measurement and interpretation

The outcome of interest is **total irrigation water use**. Suppose instead the follow-up survey asked only about **district deliveries**. That question records one source of water and omits withdrawals from private wells

If farms substitute groundwater for district water after receiving the report, district deliveries could decline while total water use remains unchanged

Gibson’s food example makes the same point: one week’s purchases omit food consumed from household stocks

Source: Gibson (2019), discussion of food purchases and consumption

Plan the sample

How many completed surveys does the study need to have enough power to detect a 2 acre-foot reduction?

What would justify a claim of an effect?

- **Test level (α):** The chance of rejecting no effect when there is no effect. Choosing $\alpha = 0.05$ sets the Type I error rate at 5%
- **Power (1 − β):** The chance of rejecting no effect when a 2-unit reduction is present. Collecting more outcomes raises power and lowers the Type II error rate β
- **Confidence interval:** Show which effect sizes are consistent with the observed difference
- **Response check:** Compare response across assigned groups to test whether missing surveys change the comparison

The test level and power target are set before data collection. The confidence interval and response check are reported after

False positives, false negatives, and power

Test the null hypothesis of **no effect of the offer** against a specified alternative: the districts' threshold of a **2 acre-foot reduction per farm-month**

Decision	**No effect is present**	**A 2-unit reduction is present**
Reject no effect	False positive (Type I error)	Detect the reduction
Do not reject	Correct non-rejection	False negative (Type II error)

At $\alpha = 0.05$, if the report has no effect, the test wrongly rejects in 5% of repeated studies. With 80% power, if the true effect is a 2-unit reduction, the test rejects in 80% of repeated studies ($\beta = 0.20$)

Source: Kim (2019), Section 2

Sample size depends on the research objective

Describing a population: If the districts only wanted to estimate average water use, the sample size would depend on how precisely the survey must estimate that mean (Lecture 5.1)

Comparing assigned groups: The districts want to know whether the offer changes water use, so the sample must be large enough to detect a specified difference between the offer and comparison groups. That calculation uses the effect size, variation in water use, test level, and power target

Sources: Lecture 5.1; Kim (2019), Section 3.5

Choosing an effect and power target

The districts set the effect of interest during the design: a reduction of **2 acre-feet per farm-month** would justify the cost of the report program

An **80% power target** is a common planning convention. If the true effect is a 2-unit reduction, the planned study would detect it in about four out of five repetitions under the assumed design. A 90% target would require more completed observations

Both choices come from the policy question and the feasible sample size, and both are fixed before any outcomes are observed. An effect size chosen after seeing an estimate no longer describes what the study was designed to detect

Source: J-PAL, Power calculations

Inputs to the power calculation

The calculation begins with the effect the study needs to detect: **2 acre-feet less water use per farm-month**

- **Outcome variation:** Standard deviation of 8 acre-feet per farm-month in each group, from pilot or prior data
- **Test:** Two-sample t test of the difference in group means, two-sided, at $\alpha = 0.05$
- **Power target:** 80%, so $\beta = 0.20$ if the true reduction is 2 acre-feet
- **Groups:** Equal numbers of farms offered the report (treatment) and not offered it (control), each farm assigned independently

Assignment by district or correlated outcomes within districts would require a calculation that accounts for clustering

Source: Kim (2019), Sections 2 and 3.5

Completed surveys needed for 80% power

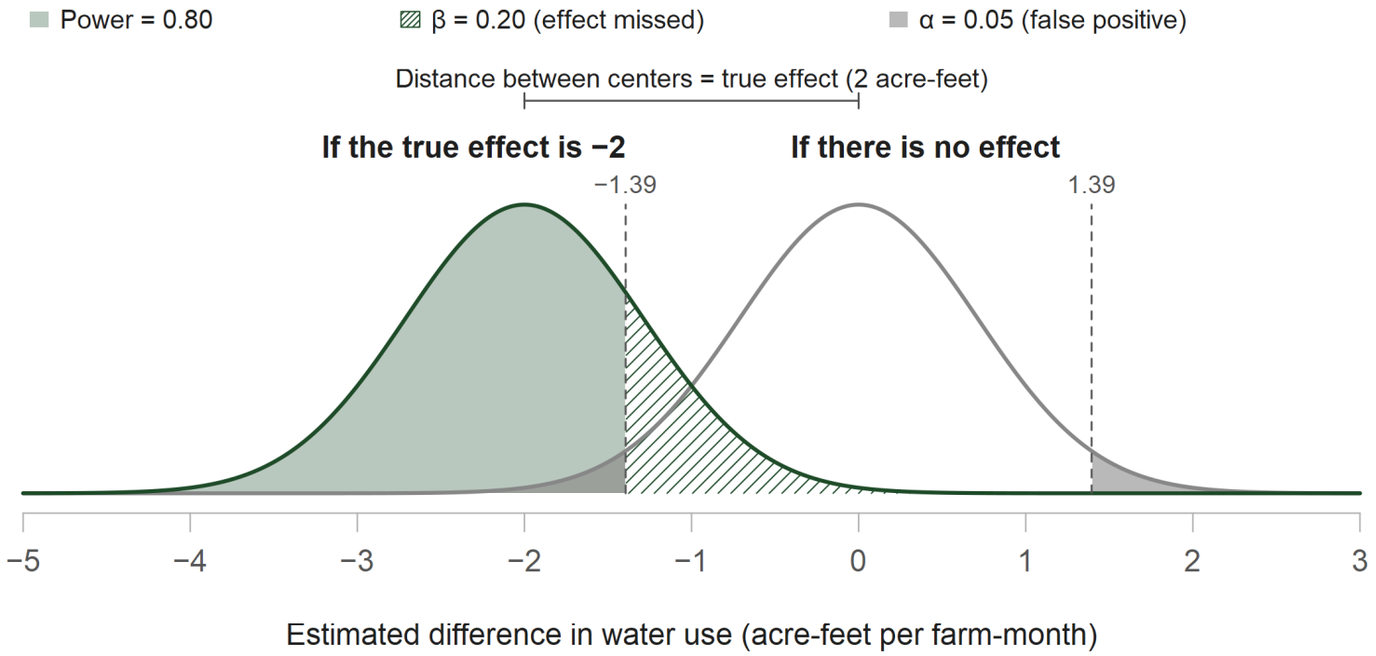
With a 2-unit effect, standard deviation of 8, and a two-sided t test at α = 0.05, an approximate two-group calculation is

$$n \approx \frac{2(8)^2(1.96 + 0.84)^2}{(2)^2} \approx 251$$

Base R's two-sample t calculation gives **252.13**, so the study plans for **253 completed surveys per group**

Illustrative calculation: `power.t.test(delta = 2, sd = 8, sig.level = .05, power = .80, type = "two.sample")`

Why 253 surveys per group give 80% power



Each curve shows where the estimate would fall across repeated studies. The spread of each curve is the standard error (0.71); the test rejects outside ±1.39 (1.96 × 0.71)

Illustrative calculation. A simulation of 10,000 repeated experiments rejects in 80.5% of runs when the true effect is −2 and in 5.0% when there is no effect

Power, β, and sample size

In the HTML version of the slides, two sliders change the number of completed surveys per group and the size of the true effect, and the chart from the previous slide updates. Selected settings:

Completed surveys per group	True effect (acre-feet)	Power	β
253	−2	0.80	0.20
51	−2	0.24	0.76
253	−1	0.29	0.71
253	−3	0.99	0.01

α is 0.05 at every setting.

Planning the survey contacts

The power calculation gives the number of **observed outcomes** needed: 253 farms per group. Because the outcome comes from a follow-up survey, the team must invite more farms than that

Suppose a pilot with similar farms suggests a **40% response rate**

$$\text{Invitations per group} = \left\lceil \frac{253}{0.40} \right\rceil = 633$$

The team would invite **633 farms in each group** (1,266 in total). This calculation checks whether the customer lists and fieldwork budget can supply the planned number of completed outcomes

The 40% response rate is an illustrative pilot assumption, not a general benchmark

What reviewers examine in a power calculation

There is no single power cutoff for every study. Reviewers examine the basis for the calculation

1. Why the specified effect size matters for the research or policy question
2. How the assumed variation in water use was estimated
3. Whether the calculation reflects assignment, clustering, and expected survey response
4. Whether the eligible farm lists can produce the required completed sample

After data collection, report the realized sample and the precision of the estimated effect

Sources: J-PAL, Power calculations; course empirical strategy checklist

Multiple testing and false positives

The districts may also want to know whether the report changes irrigation scheduling and crop choice. Testing **three outcomes** creates three opportunities to reject a true null hypothesis

If the three tests were independent and each used a 5% level, the chance of at least one false positive would be

$$1 - (1 - 0.05)^3 \approx 0.14$$

The outcomes may be correlated, so this value is illustrative. Before analysis, identify the primary outcome, then report all tested outcomes and explain any adjustment for multiple comparisons

Source: Kim (2019), discussion of multiple testing

Interpret the results

What do the data support once they arrive?

Random variation and survey response

Farms were randomly assigned to the report or comparison group. The follow-up survey records water use only for farms that respond. Before interpreting the difference between groups, ask

1. **Random variation:** Could another assignment produce a different group difference? A confidence interval or test quantifies this uncertainty
2. **Selective response:** Do respondents differ from farms that did not answer? Compare response rates and available farm characteristics by assigned group

If survey response is 8%

At the planned **40% response rate**, 633 invitations would yield about **253 completed surveys per group** and 80% power under the stated assumptions

Suppose the realized response rate is **8%**. The same invitations yield about **51 completed surveys per group**, and power to detect the 2-unit effect falls to about **24%**

To obtain 253 completions at an 8% rate, the team would need about **3,163 invitations per group**. The customer lists may contain fewer eligible farms

Illustrative calculation: R `power.t.test(n = 51, delta = 2, sd = 8, sig.level = .05, type = "two.sample")`

Nonresponse and interpretation

There is no universal expected response rate. Use a pilot or comparable survey to plan invitations. An 8% rate calls for evidence about who responded

- Compare respondents with the customer lists on available farm characteristics
- Report response rates separately for the offer and comparison groups
- Examine whether follow-up or weighting changes the estimate

Selective response can bias the estimated effect if it changes the comparability of the assigned groups. A larger invitation list alone does not resolve that concern

Source: AAPOR, *Response Rate Overview*

Precision after an 8% response rate

With about **51 farms per group**, suppose the estimate is $\widehat{\tau} = -1.5$ acre-feet per farm-month and the standard deviation is 8 in each group. The approximate standard error is

$$SE \approx \sqrt{\frac{8^2}{51} + \frac{8^2}{51}} = 1.58$$

An approximate 95% interval is **−1.5 ± 1.96(1.58) = [−4.6, 1.6]** acre-feet. The data remain compatible with no effect and with reductions larger than the 2-unit threshold set during planning

The small completed sample explains both the fall to 24% power and the wide interval. The interval describes sampling uncertainty among respondents; it does not address selective response

Why consider an equivalence test?

The estimate is not statistically distinguishable from zero, and the districts are considering ending the report program

The 95% interval **[−4.6, 1.6]** still includes reductions greater than 2 acre-feet. The study has not ruled out a benefit large enough to justify the program

To support a claim that the effect is negligible, the team must specify what size of change would matter and test whether the data rule out effects that large

Equivalence testing addresses that question

Source: Kim (2019), Section 3.7

Equivalence test for the water-use report

1. Before analysis, define **−1 to 1 acre-foot** as a negligible change in monthly use
2. For a 5% equivalence test, calculate a **90% confidence interval**. The entire interval must lie inside the negligible range
3. With 51 respondents per group, the approximate interval is **[−4.1, 1.1]** acre-feet

The interval extends below −1. The data leave open reductions larger than 1 acre-foot, so they do not establish a negligible effect

Source: Kim (2019), Section 3.7

Takeaways

Topic	Question to consider
Population and sampling frame	Which units do the data cover, and which do they omit?
Outcome measure	How does the recorded measure relate to the concept in the research question?
Comparison	Which groups are compared, and what assumption supports the comparison?
Sample size	If collecting data, what effect size matters, and how many observations would detect it?
Primary outcome	If testing several outcomes, which one is primary?
A null result	Which effect sizes would the confidence interval need to rule out?

See also: course empirical strategy checklist

Check-in

Report from Tuesday’s methods workshop

Students who did not receive feedback on Tuesday can report their current methods plan

References

Falk, Armin, and James J. Heckman. 2009. “Lab Experiments Are a Major Source of Knowledge in the Social Sciences.” *Science* 326 (5952): 535–38. https://doi.org/10.1126/science.1168244.

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